Do Now · retrieval

OCR pseudocode, files and IDEs

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

If y=4 is passed by ref and a method adds 5, what is y after the call?

Q2

What does a breakpoint do?

Q3

Which condition controls a while loop?

Q4

If a file has two lines and each readLine reads one line, how many calls read the whole file?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Open the resource — openRead returns a file object for reading; it does not overwrite the file.

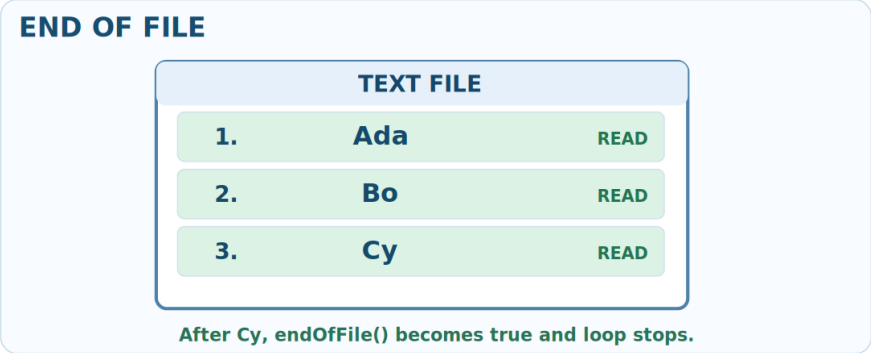
Core idea 2

Read until EOF — endOfFile() is false while another line remains. readLine() retrieves one line and moves the file position.

Core idea 3

Close the file — After the final line, endOfFile() is true, so the loop stops and the file is closed. For writing use openWrite , writeLine , then close() ; unlike reading, opening for writing overwrites any existing file contents .

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

openRead — Open a file for reading using OCR A Level pseudocode.

readLine — Read the next line from an open file.

endOfFile — Test whether no unread lines remain.

openWrite — Open for writing; existing file contents are replaced.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Trace OCR pseudocode that reads the first two lines of fruits.txt.

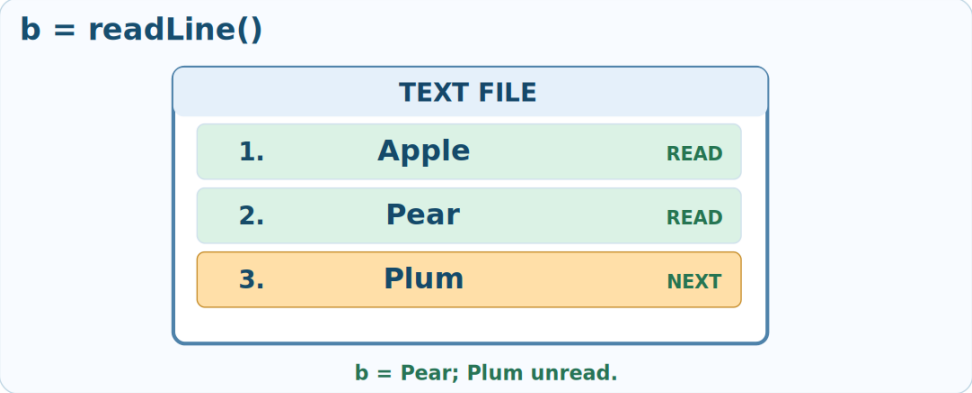
Follow the worked method

1. Predict first — The first call returns Apple and advances the file position.

2. Model and check — close releases the file resource; Plum has not been read.

3. Predict a later read — f.close() releases the file. A separate later openRead("scores.txt") would read Ada,82 then Ben,74 in that order.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
The loop prints York from the first readLine call.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

An OCR pseudocode file contains Cat, Dog, Fox on separate lines. What do the first two readLine() calls return?

Question 2

Write OCR A Level pseudocode to open words.txt, display each line and close the file.

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain why openWrite is unsuitable for reading an existing file. Write OCR pseudocode that replaces events.txt with one line HELLO, and identify a debugging step if a reading loop omits its last record.

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

OCR's openWrite("data.txt") safely reads existing data without altering it.

Use key vocabulary from memory

1. Explain openRead in your own words.

2. Explain readLine in your own words.

3. Explain endOfFile in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.